

In the figure, the value y is obtained using the formula: $y = V_0 \sin \theta t \pm \frac{1}{2} g t^2$

Substituting the value of the time of flight " t " in the above equation:

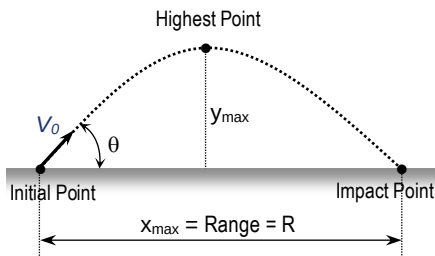
$$y = V_0 \sin \theta \left(\frac{x}{V_0 \cos \theta} \right) \pm \frac{1}{2} g \left(\frac{x}{V_0 \cos \theta} \right)^2$$

Simplifying the above equation will produce the "**General Equation of Projectile**".

$$y = x \tan \theta - \frac{g x^2}{2 V_0^2 \cos^2 \theta}$$

Since the variable " x " is in second degree while the other variable " y " is in the first degree, then the graph is a **parabola**.

Note: The vertical component of the velocity decreases as it goes up and is zero at maximum point of the projectile and increases as it goes down, while the horizontal component is constant. If the initial point of the projectile is on the same elevation as the point of impact, the following formulas may be used:



Maximum height, $y_{\max}$: $y_{\max} = \frac{(V_0 \sin \theta)^2}{2g}$

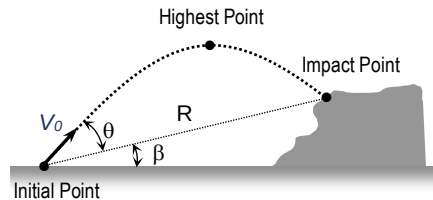
Range, $x_{\max}$ or R : $R = \frac{V_0^2 \sin 2\theta}{g}$

In the absence of air drag, the following rules apply to the case of a level target:

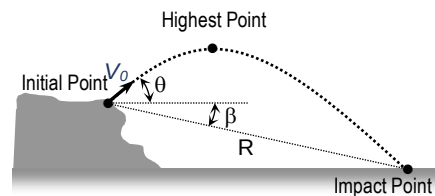
1. The trajectory is parabolic.
2. The impact velocity is equal to initial velocity.
3. The impact angle is equal to the initial launch angle.
4. The range is maximum when $\theta = 45^\circ$.
5. The time for the projectile to travel from the launch point to the apex is equal to the time to travel from apex to impact point.

6. The time for the projectile to travel from the apex of its flight path to impact is the same time as initially stationary object would take to fall a distance of maximum altitude.

If the initial point of the projectile is not on the same elevation as the point of impact, the following formulas for range, R may be used:



$$R = \frac{2 V_0^2 \sin \theta \cos(\theta + \beta)}{g \cos^2 \beta}$$

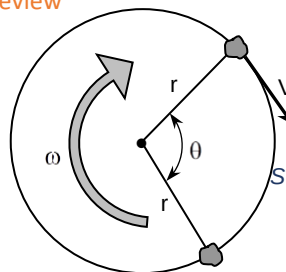


$$R = \frac{2 V_0^2 \sin \theta \cos(\theta - \beta)}{g \cos^2 \beta}$$

Rotational Motion

Rotation is a motion along a circular path.

Excel Review



Let: S = linear distance in meters
 V = linear velocity in m/s
 a = linear acceleration in m/s^2
 θ = angular distance in radians
 ω = angular distance in rad/s
 α = angular acceleration in rad/s^2

$$S = r\theta \quad V = r\omega \quad a = r\alpha$$

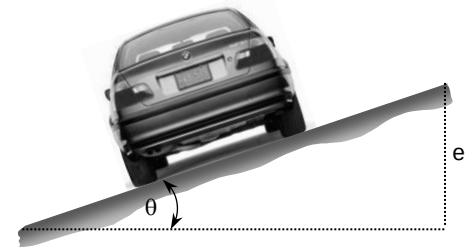
For variable angular acceleration, the following formulas must be used.

$$\omega^2 = \omega_0^2 \pm 2\alpha\theta \quad \omega = \omega_0 \pm \alpha t$$

$$\theta = \omega_0 t \pm \frac{1}{2} \alpha t^2$$

Banking of Curves

In order for the weight of the car to help the frictional force resist the centrifugal force, banking of highway curve is necessary. Once banked, a car can move at higher speed due to bigger resistance of the centrifugal force.



The ideal angle of banking, θ may be calculated using:

$$\tan \theta = \frac{V^2}{gr}$$

For greater velocity without skidding,

$$\tan(\theta + \phi) = \frac{V^2}{gr}$$

$$\tan \phi = \mu$$

where:

ϕ = angle of friction

μ = coefficient of friction

The superelevation, e , is the rise in the outer edge of the roadway.

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